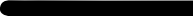
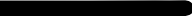


APACHE KYLIN

By Reynard A.



ENVIRONMENT SETUP

Setup Docker Desktop, Kylin, Kafka, and Related Environment

Step 1 : `docker pull apachekylin/apache-kylin-standalone:3.0.1`

Step 2 : `docker run -d --name kylin --hostname kylin -m 8G -p 7070:7070 -p 8088:8088 -p 50070:50070 -p 8032:8032 -p 8042:8042 -p 16010:16010 apachekylin/apache-kylin-standalone:3.0.1`

Step 3 (For Windows) :

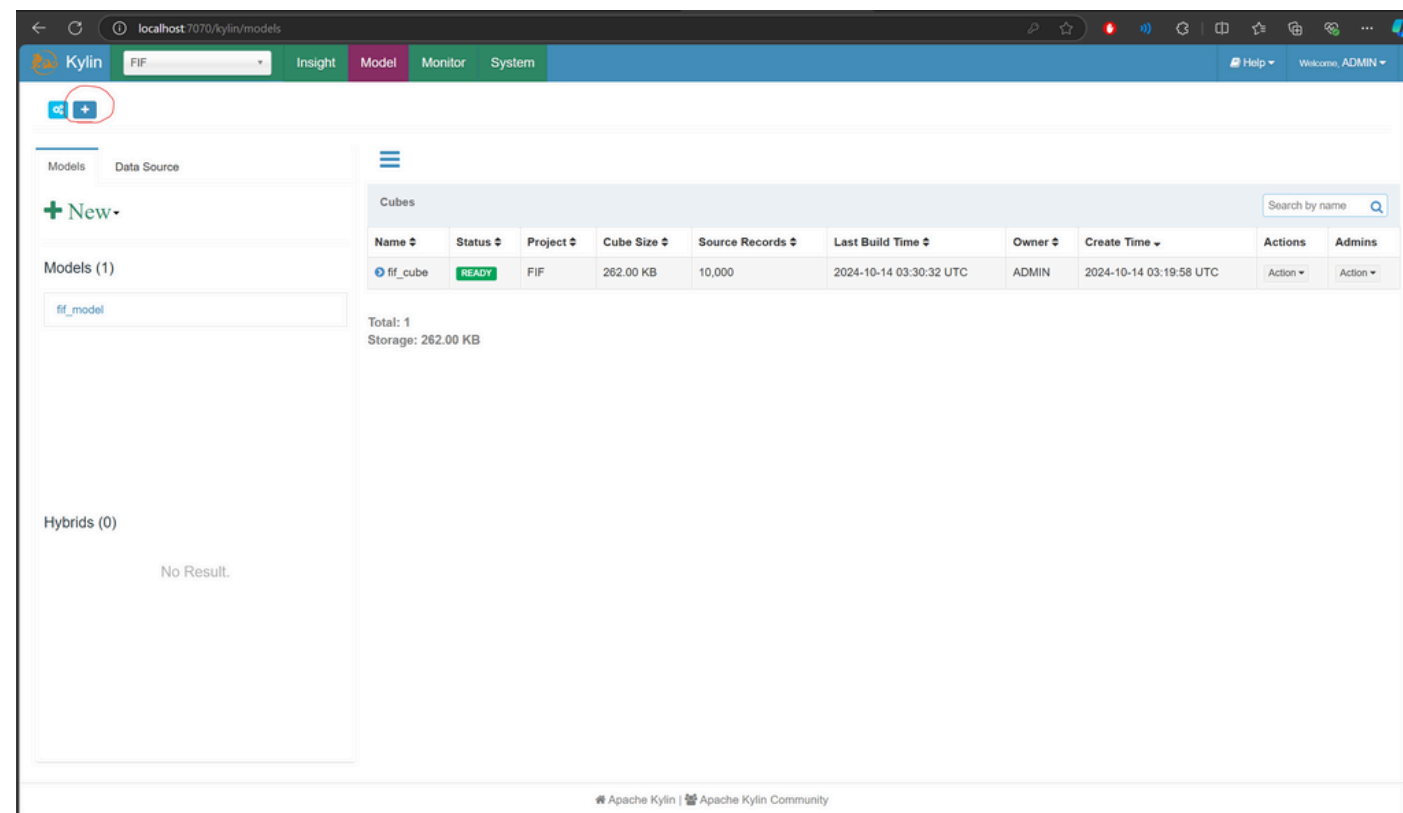
Make file in `C://users/[your user name]/.wslconfig`

content:

`[wsl2]`

`kernelCommandLine = vsyscall=emulate`

MAKE NEW PROJECT



Click '+' button in top left to Create Project

New Project

Project Name

You can use letters, numbers, and underscore characters '_'

Project Description

Project Description..

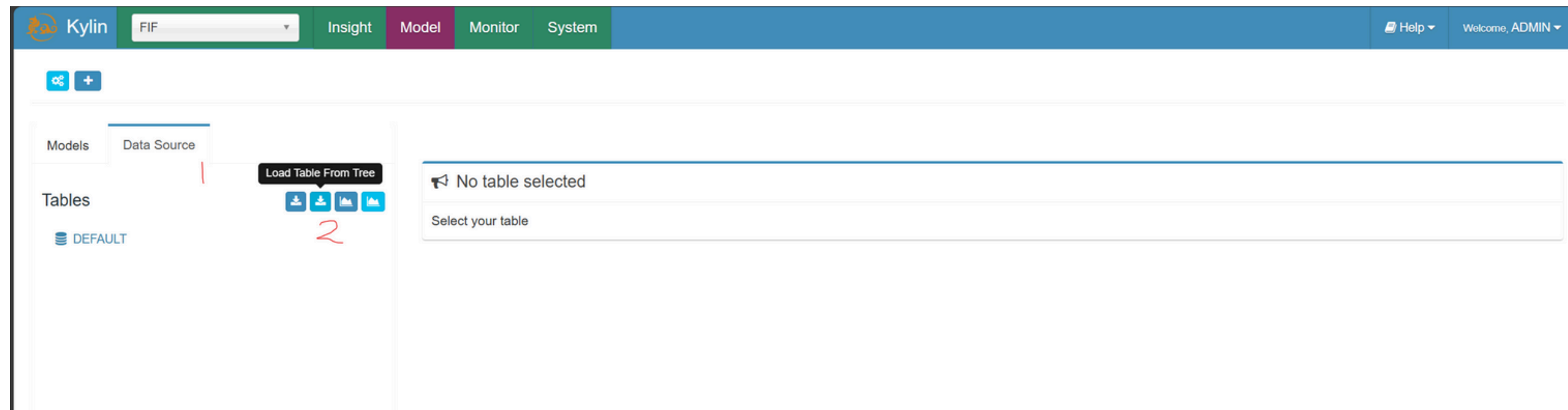
Project Config

+ Property

Close Submit

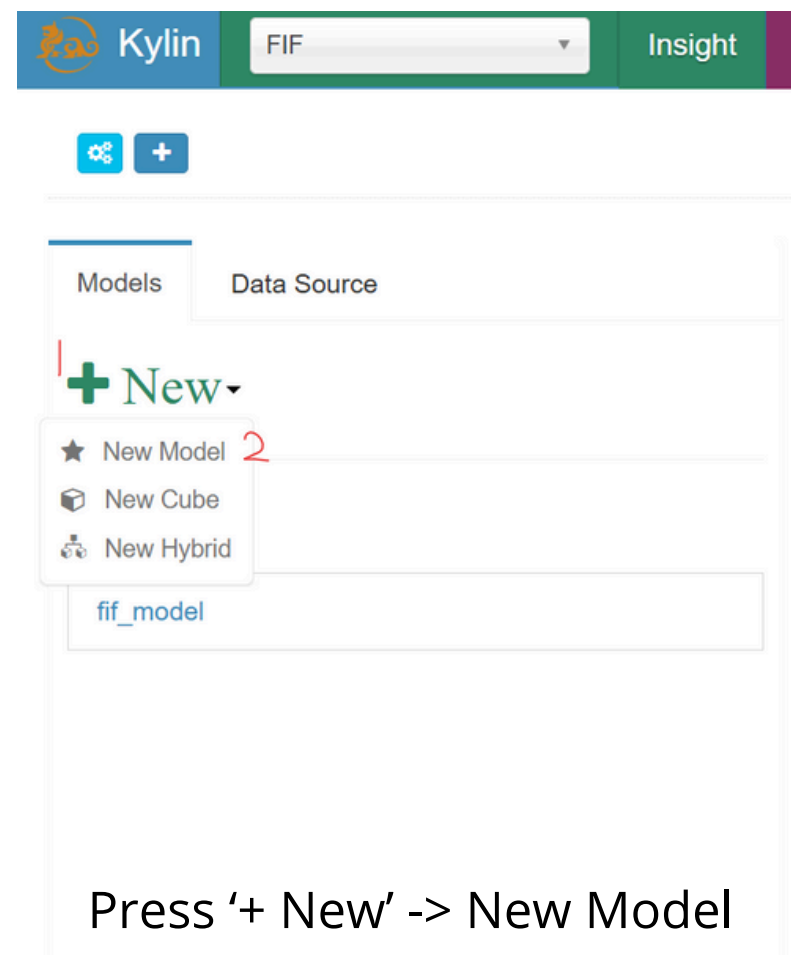
Add project name (FIF) and description and config if necessary

MAKE PROJECT

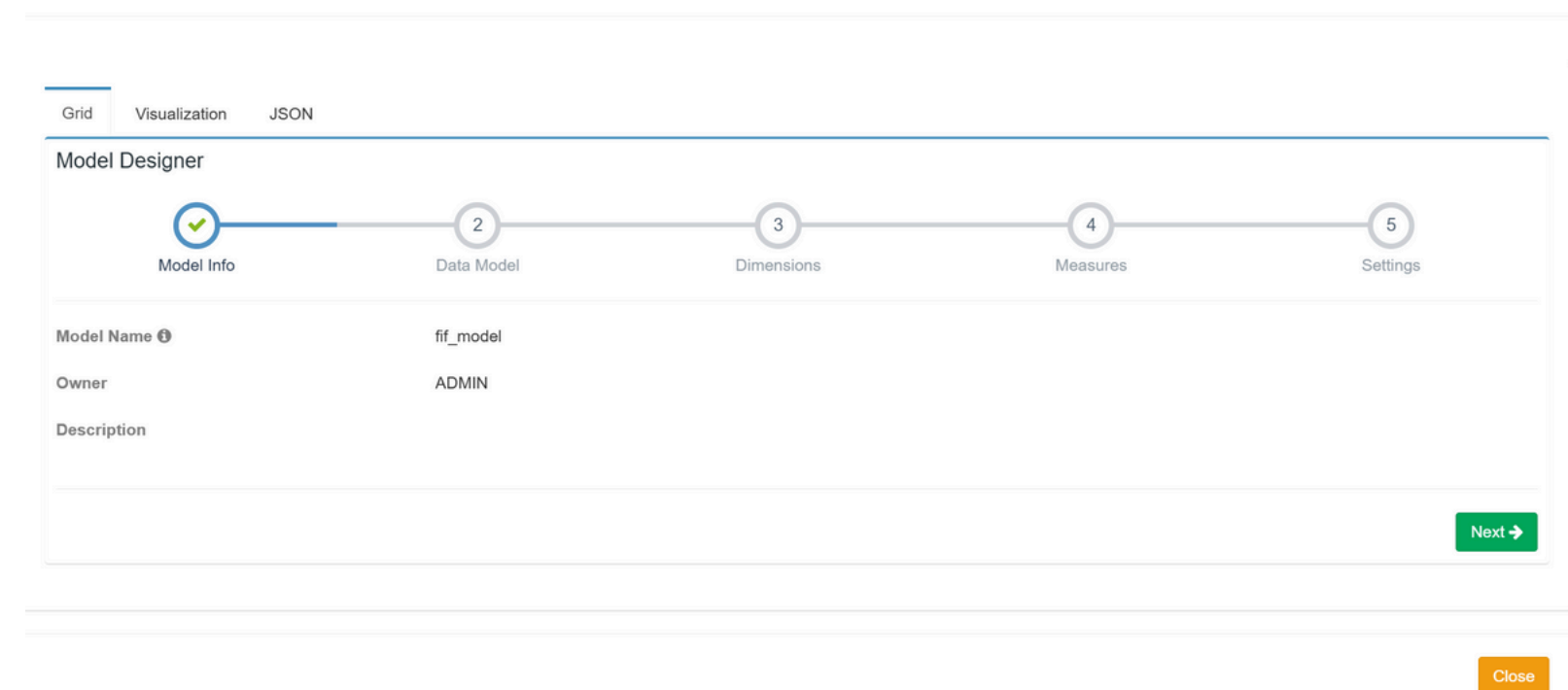


Go to Data Source -> Load Table From Tree -> Select desired table source

MAKE NEW MODEL

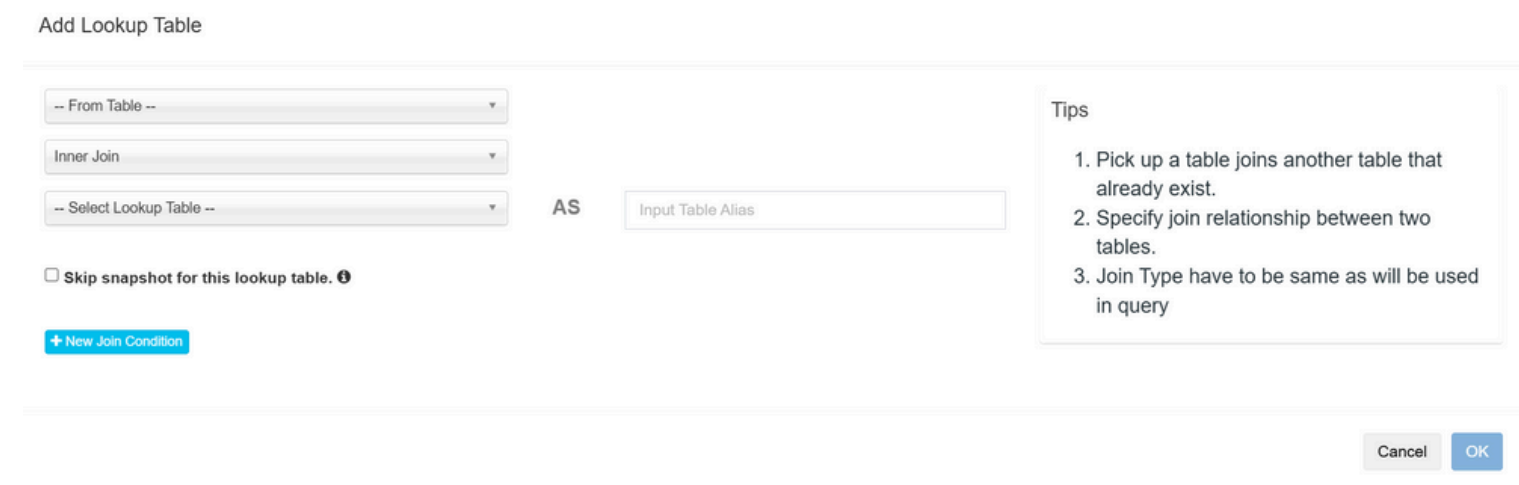
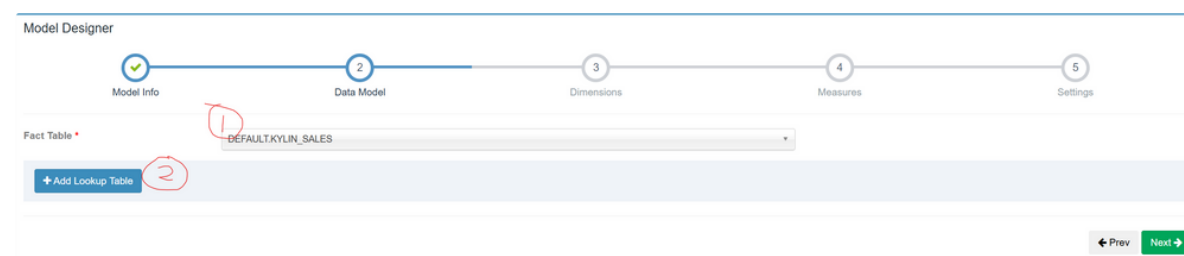


Press '+ New' -> New Model



Step 1 (Model Info) : Enter Model Name (fif_model)

MAKE NEW MODEL



Step 2 (Data Model):

Select Fact Table (Measured) and Add Lookup Table for joining to other table

In this case we want to get SUM Price with buyer based on country.

So:

We want to make KYLIN_SALES as the fact, to get the country we need to get the country we join
KYLIN_SALES.BUYER_ID with KYLIN_ACCOUNT.ACCOUNT_ID ->

Join KYLIN_ACCOUNT.ACCOUNT_COUNTRY with KYLIN_COUNTRY.COUNTRY

MAKE NEW MODEL

Model Designer

Model Info Data Model Dimensions Measures Settings

Select dimension columns

ID	Table Alias	Columns
1	KYLIN_SALES	BUYER_ID ✕
2	KYLIN_ACCOUNT	ACCOUNT_ID ✕ ACCOUNT_COUNTRY ✕
3	KYLIN_COUNTRY	COUNTRY ✕

← Prev Next →

Step 3 (Dimensions) :
Select the desire dimension columns for each table alias

MAKE NEW MODEL

Model Designer

Model Info Data Model Dimensions Measures Settings

Select measure columns

ID	Table Alias	Columns
1	KYLIN_SALES	PRICE ✕

← Prev Next →

Step 4 (Measures) :
Select the desired measure columns

MAKE NEW MODEL

Model Designer

Model Info Data Model Dimensions Measures Settings

Partition

Partition Date Column ⓘ

--Select Partition Table-- --Select Partition Column--

Date Format

yyyy-MM-dd

Filter

Filter ⓘ

WHERE

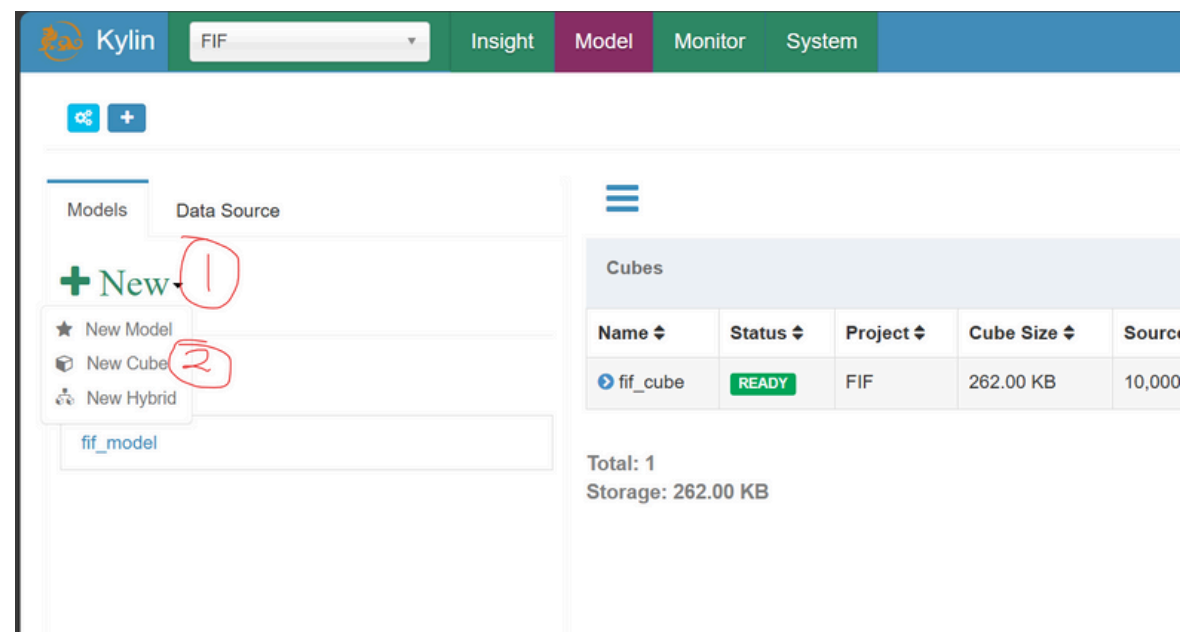
Please input WHERE clause without typing 'WHERE'

← Prev Save

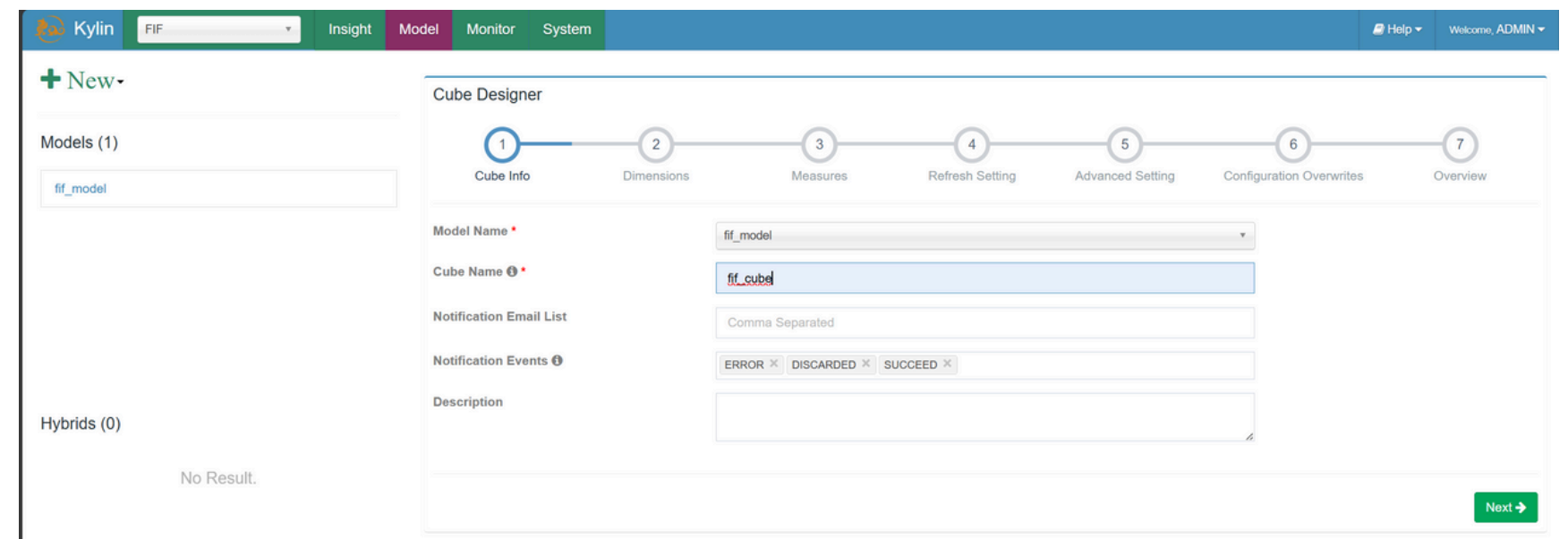
Step 5 (Settings) :

Add additional setting if necessary (partition is recommended for better performance and optimized data)

MAKE NEW CUBE

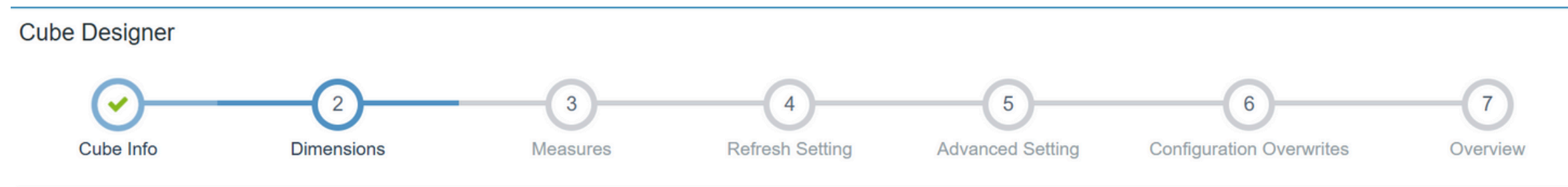


Make New Cube '+ New' -> 'New Cube'



Step 1 (Cube Info) : Fill cube general info (Model Name, Cube Name , etc)

MAKE NEW CUBE



Step 2 (Dimensions) : Select Dimensions: Choose the attributes that describe your data, such as BUYER_ID, ACCOUNT_ID, ACCOUNT_COUNTRY, and COUNTRY

Step 3 (Measures) : Identify and input numerical data that you want to analyze, such as PRICE, ITEM_COUNT, or any aggregated calculations (e.g., total sales). In this case is SUM to make TOTAL_PRICE

Step 4 (Refresh Settings) : Set how often you want the data to be refreshed (e.g., daily, weekly).

Step 5 (Advanced Settings) : Decide if you want to partition your data (e.g., by date) for performance optimization.

Step 6 (Configuration Overwrites) : Identify any specific settings that should override the default configurations (e.g., specific partition settings or aggregation methods).

Step 7 (Overview) : Provide a summary of all the previous steps, highlighting key dimensions, measures, and settings.

MAKE NEW CUBE

The screenshot shows the Kylin web interface at localhost:7070/kylin/models. The 'Model' tab is selected, indicated by a red circle '1'. The interface displays a table of cubes with the following data:

Name	Status	Project	Cube Size	Source Records	Last Build Time	Owner	Create Time	Actions	Admins
fif_cube	READY	FIF	262.00 KB	10,000	2024-10-14 03:30:32 UTC	ADMIN	2024-10-14 03:19:58 UTC	Action	Action

The 'Actions' dropdown menu for the 'fif_cube' row is open, showing options: Edit, Build, Refresh, Merge, Lookup Refresh, Disable, and Clone. The 'Build' option is highlighted with a red circle '3'. The 'Admins' column shows an 'Action' dropdown for the 'fif_model' entry.

Finally Build The Cube By Clicking based on the Figure Above

RESULT

Kylin FIF Insight Model Monitor System Help Welcome, ADMIN

Tables
DEFAULT

New Query Saved Queries Query History

```
1 SELECT KYLIN_COUNTRY.COUNTRY, SUM(KYLIN_SALES.PRICE) AS TOTAL_PRICE
2 FROM KYLIN_SALES
3 JOIN KYLIN_ACCOUNT ON KYLIN_SALES.BUYER_ID = KYLIN_ACCOUNT.ACCOUNT_ID
4 JOIN KYLIN_COUNTRY ON KYLIN_ACCOUNT.ACCOUNT_COUNTRY = KYLIN_COUNTRY.COUNTRY
5 GROUP BY KYLIN_COUNTRY.COUNTRY
6 ORDER BY TOTAL_PRICE DESC;
```

Project: FIF LIMIT 50000 Submit

Results
1 x

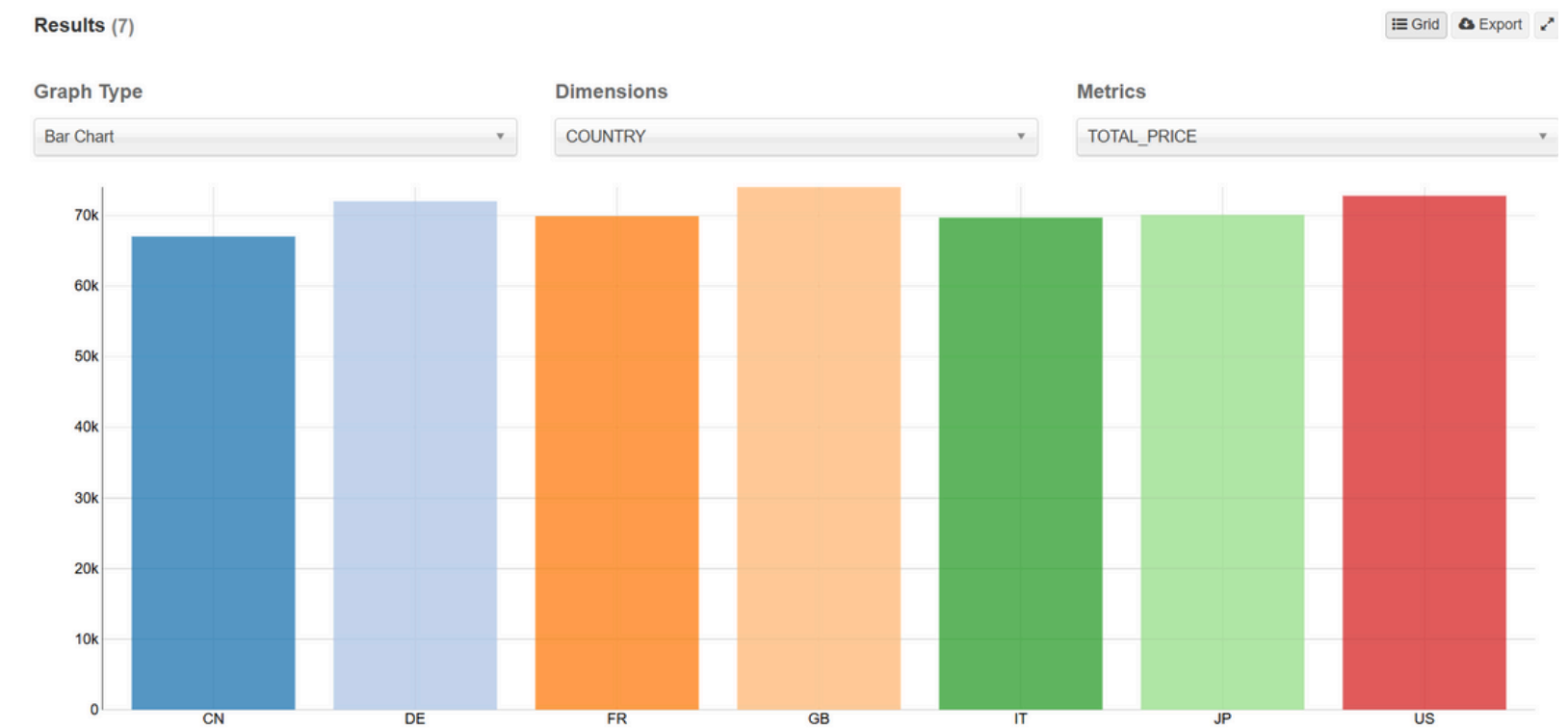
Query String Start Time: 2024-10-14 04:40:40 UTC Duration: 0.65s Return Save

Status: Success Project: FIF Cubes: CUBE[name=fif_cube]

Results (7)

COUNTRY	TOTAL_PRICE...
GB	74052.1557
US	72807.8659
DE	72034.5262
JP	70119.6708
FR	69939.0301

Apache Kylin | Apache Kylin Community



To Show the result, go to insight -> enter the desired SQL Query -> result can also be shown in graph by pressing 'Visualization'



THANK YOU